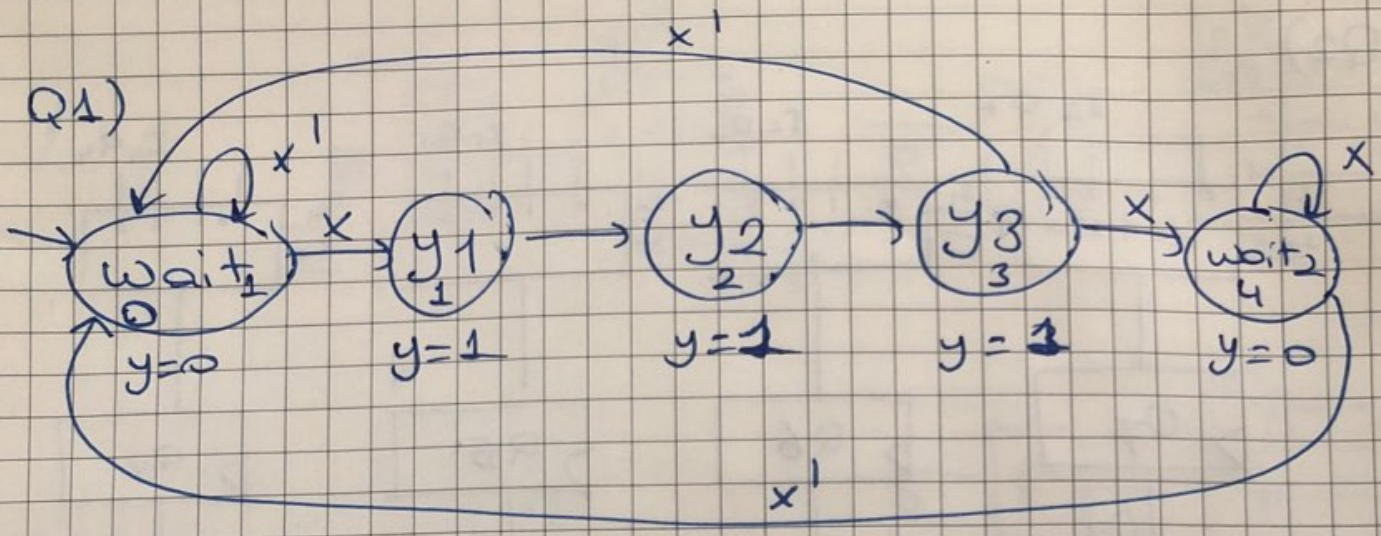
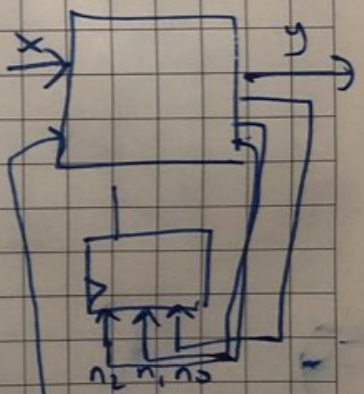
Q1)



Truth table

S_2	S_1	S_0	x	n_2	n_1	n_0	y
0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0
0	0	1	x	0	1	0	1
0	1	0	x	0	1	1	1
0	1	1	0	0	0	0	1
0	1	1	1	1	0	0	1
1	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0



$n_2 = S_2' S_1 S_0 x + S_2 S_1' S_0' x$

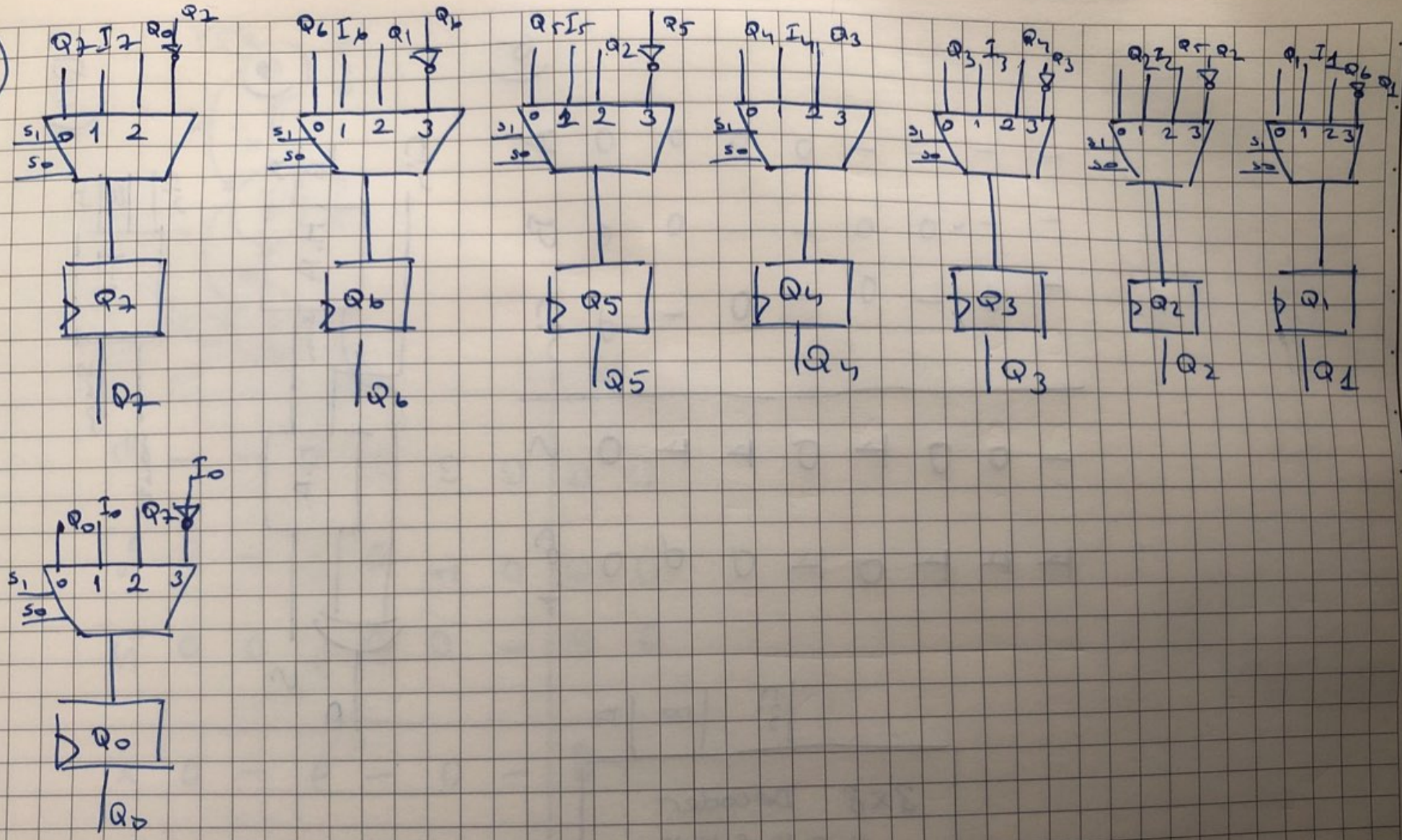
$n_1 = S_2' S_1' S_0 + S_2' S_1 S_0'$

$n_0 = S_2' S_1' S_0' x + S_2' S_1 S_0$

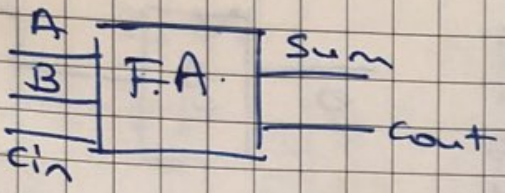
$y = S_2' (S_1' S_0 + S_1 S_0' + S_1 S_0)$

morning glory

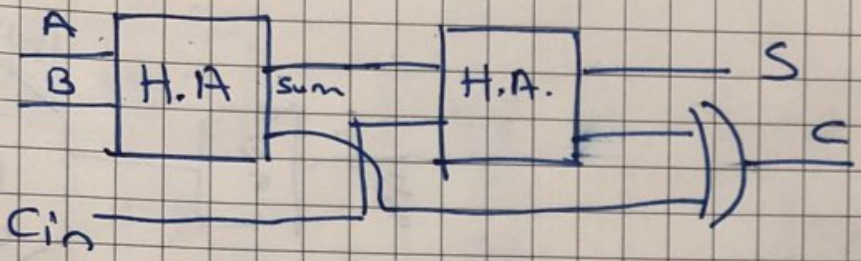
2)


Q7 I7 Q6 Q7
S1 S0 0 1 2
Q7
Q7
Q6 I6 Q5 Q6
S1 S0 0 1 2 3
Q6
Q6
Q5 I5 Q4 Q5
S1 S0 0 1 2 3
Q5
Q5
Q4 I4 Q3
S1 S0 0 1 2 3
Q4
Q4
Q3 I3 Q2 Q3
S1 S0 0 1 2 3
Q3
Q3
Q2 I2 Q1 Q2
S1 S0 0 1 2 3
Q2
Q2
Q1 I1 Q0 Q1
S1 S0 0 1 2 3
Q1
Q1
Q0 I0 Q7 I0
S1 S0 0 1 2 3
Q0
Q0

Q3)

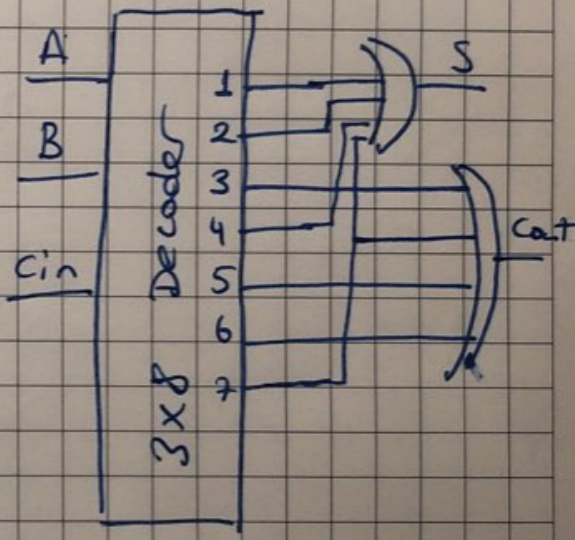


a)



b)

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Q4)

S_1	S_0	X	S_1'	S_0'	Z
0	0	0	0	0	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	0	0	0
1	0	0	0	1	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	1	0	0	0

X

